

Manoj Sai Penugonda

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SUMMARY

Data Scientist with 3.6 years of experience in building scalable Machine Learning and GenAI solutions. Strong expertise in data processing, feature engineering, predictive modeling, applied GenAI use cases, prompt engineering, transforming raw data into actionable insights that improve business outcomes and collaborating effectively with stakeholders for strategic decision-making.

EDUCATION

Anna University
Madras Institute of Technology

08/2018 - 07/2022
B.E. in Electronics Engineering

EXPERIENCE

Data Scientist
Accenture

07/2022 - Present
Chennai

Data Profiling

- Profiled over 5000 records to identify and resolve 1500+ data quality issues, improving **data accuracy** by **15%**.
- Performed comprehensive **data profiling** on multiple datasets, detecting inconsistencies and discrepancies, and providing **actionable insights** that reduced data cleaning time by **20%**.
- Analyzed and documented **key data patterns**, enabling better data organization and improving data usability by **25%**.

ML Based Synthetic Data Generation Bot

- Engineered an ML-driven bot that generated over 1000+ rows of data while maintaining 95% fidelity to the original dataset's **statistical relationship, correlation, and distributions**.
- Built a **regression-based ML model** that achieved **80% accuracy** in predicting weekly product sales volume based on promotional data and historical trends.
- Integrated the synthetic data generation pipeline with downstream analytics tools, accelerating model training by **25%** and **reducing overfitting** by **20%**.
- Designed and implemented a data augmentation pipeline using statistical sampling and generative models, increasing dataset diversity by **30%** and improving **model generalization** on unseen data.

Text Similarity Search - RAG Framework

- Developed a **Retrieval-Augmented Generation (RAG)** based chatbot leveraging **Hugging Face Embedding Model** (thenlper/gte-small) and Vector Database (**ChromaDB**), to improve search accuracy by over **25%** using **cosine similarity** techniques.
- Executed semantic similarity search on historical text data, comparing user input against past records and retrieving **top-K** matches with **95%** precision.
- Led a **GenAI** PoC for the conversion of user stories into test cases, increasing the project pipeline efficiency by **30%**.
- Optimized vector search efficiency by **20%** through refined similarity search parameters and improved data preprocessing, enhancing retrieval speed and relevance for large-scale datasets.
- Created a user-friendly GUI with **Streamlit**, reducing decision-making time for non-technical stakeholders by over **40%** and significantly improving accessibility.

TECHNICAL SKILLS

Programming and Query Languages: Python, SQL, React.js

Machine Learning & Deep Learning: Linear & Logistic Regression, Decision Trees, Random Forest, Gradient Boosting, AdaBoost, Neural Networks (ANN, CNN, RNN), Transformers, Generative AI (LLM Fine-Tuning -LoRA, GANs), Diffusion Models, NLP, Clustering, Optimization Techniques

Data Science Libraries: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, TensorFlow

Tools & Technologies: Databricks, Azure, Git, GitHub, RESTful APIs, Fast API

Analytics: Statistical Analysis, Data Visualization, Data Wrangling, Feature Engineering, A/B Testing, Hypothesis Testing

CERTIFICATIONS

- GH-300 GitHub Copilot.
- Certificate Programme in Data Science and Machine Learning — IIT Delhi.